

Chapter – 2: Understanding the Weather

A change in the weather is sufficient to create the world and oneself anew. – Marcel Proust (French novelist)

Weather and its Element

- Winter morning – wake up and shiver – wear warm clothes
- Summers – wear cool, comfortable clothes

What is weather?

- **Weather** – state – earth's atm. – particular place, time
- Simple terms – atm. – layer of gases – surround some planets – case of earth – called air
- Earth's atm. – comparable to cake – several layers
- Layer – closest to earth's surface – troposphere – all land-based plants, animals – live here
- This layer – also – all weather phenomenon – happens here
- Troposphere – 6-18 kms – less thick at poles – thicker in tropical zone
- Use many words – describe weather – hot, cold, rainy, cloudy, humid, snowy, windy, etc.
- They describe – diff. ways – experience elements of weather
- Elements of weather –
 - **Temperature** – how – hot/cold – atm.
 - **Precipitation** – any form – water – rain, snow, sleet, hail – falling from sky
 - **Atmospheric pressure** – weight of air – felt on surface
 - **Wind** – movement of air – speed, dir.
 - **Humidity** – amt. – water vapour in air
- Krishnan (Chennai) – speaking to – Amir (Kashmir) – Amir asks him – how cold – Krishnan tell him its chilly – confusion
- People need – commonly agreed way – measure temp., other elements
- From early times – humans – obs. nature – learnt to read – signals – forecast weather
- Birds flying low – ant carrying eggs – squirrels gathering nuts, etc. – provide info. – rain, storms
- This knowledge – passed – gen. to gen.
- Today even – many parts – India – people use – traditional method – predict weather

Observing nature's clue

- Ants – shift eggs – higher grounds – indicates – expected change in weather – heavy rain
- Frogs – croaking – forest – western ghats – expects rain
- Pine cones – natural mechanisms – open in dry weather – close in humid weather
- Last few centuries – scientists – worked out methods – measure, monitor elements – great precision
- Based on inputs – meteorologists – predict how weather will react in some time

Weather Instruments

Temperature

- Thermometers – several types – some measure – ambient temp. – others – max. and min. temp.
- Thermometers – often use – coloured liquid – expands with increasing temp.
- BUT – digital thermometers – preferred – more precise – record more data
- Temp. recordings – collect useful statistics –

- **Range of Temp.** – max. temp. – min. temp. – during a day
- **Mean daily temp.** – max. temp. + min. temp. divided by 2

Precipitation

- News says – particular place – 30 mm rainfall – how to measure?
- Amt. of rainfall – measured – instrument – **rain gauge**
- During rain – water falls into funnel – collected in cylinder – scale attached to cylinder – measure depth – rainwater

Atmospheric pressure

- Our bodies – aware – temp., rainfall – BUT – sometimes – weather feels heavy
- This phenomenon – related to – atm. pressure – weight of air – above, around us
- Atm. pressure – high – sea coast – gets low – go higher – mountains
- Climb mountain – air – thinner – less oxygen – body – work harder – people feel dizzy
- Sometimes – atm. pressure – drop suddenly – plain areas – meteorologists call it – low-pressure system – develops into – storm, cyclone
- Instrument – measure atm. pressure – **barometer** – several types
- Unit used – generally – **millibar** (mb)
- Normal atm. pressure – sea coast – 1013 mb – pressure below 1000 mb – depression

Wind

- Movement of air – high pressure areas to low pressure areas
- 2 imp. factors – speed, dir. – describe wind
- Wind – imp. element – dir., speed – help – weather forecasting
- Air pilots, sailors – must be aware – wind data – wind – great influence – flying, sailing
- Farmers – use wind dir. – predict rain – greater speed – soil – dry faster
- Wind vane – rotating arm – pointer – one end – tail – other end
- Wind blows – tail is pushed – pointer turns in dir. of wind
- Wind vane on tarmac – wind sock – indicate pilot – dir. of wind – take-off, landing
- Similar socks – used – industries – release gases
- Simplest instrument – measure – speed, dir. – **anemometer**
- Anemometer – 3, 4 metal cups – rotate on vertical shaft – stronger wind – faster rotation
- Meter – attached at bottom – counts rotation – calculates speed – km/hr

Humidity

- Last element – our list – humidity – amount of water vapour – in air
- Depends on factors – temp., wind, pressure, location
- Humidity of air – **rel. humidity** – air – no water vapour – 0% – air – saturated – water vapour – 100%
- Dry weather – rel. humidity – 20% to 40% - humid weather – 60% to 80%
- Measure humidity – instrument – **hygrometer** – several types
- Measurement – humidity – very imp. – food processing, museums

Weather Stations

- Several instruments – measure weather – particular place
- Weather station – all these instruments – together – easier – track weather

- Readings – all measurement – taken at regular intervals – mapping, forecasting weather

Automated weather station

- AWS – self-operating sys. – various sensors – measure weather data – temp., humidity, wind speed, dir., precipitation, atm. pressure
- Such stations – widely used – agriculture, aviation, navigation, env. monitoring, etc. – provide accurate info. – without human intervention

Predicting the Weather

- Meteorologists – collect data – long periods – study data – use scientific methods – predict weather
- Such predictions – very imp. – climate change – extreme weather – droughts, floods, cyclones, etc. – more frequent
- Accurate predictions – helps – be ready – such events
- Alos enable – local govt. – mobilise resources – prepare – disasters
- Stormy weather – expected at sea – fishermen – warned – venture out
- Cyclone – expected – entire coastal area – evacuated
- This map – India Meteorological Department – 19th May 2024

